

Project Announcement

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Project

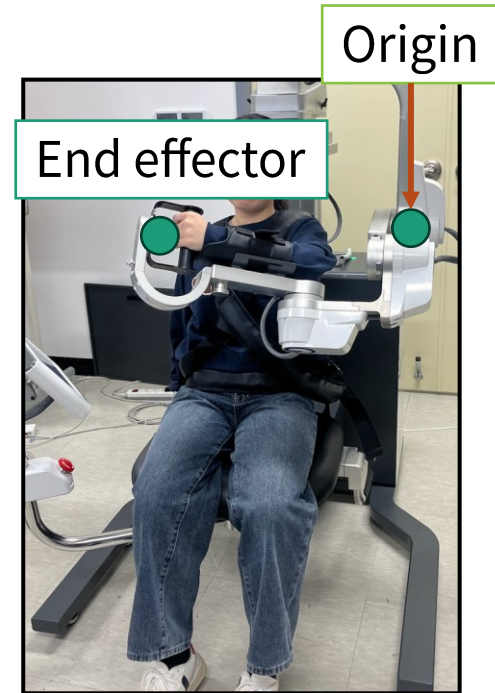
- 50% of total marks
- IC-PBL M-type Project
- Topic: Human motion(trajjectory) classifier
 - Basic dataset will be given on ~~10~~11th week.
 - Then, the sub-quests will be:
 - 1) Augment or refine the dataset by yourself.
 - 2) Build a model that **classify the input into the given categories**
 - 3) Suggesting an evaluator of your own model with self-set error metrics and proving its validity.

Dataset

- In the dataset, each row contains several columns separated by **commas (,)**. Each column represents different information, such as motor angles, speed, current, and sensor readings.
- For our analysis, **you only need to use column [6]**, which represents the end effector position.
 - (Since the column index starts from 0, this is actually the 7th column in each row.)

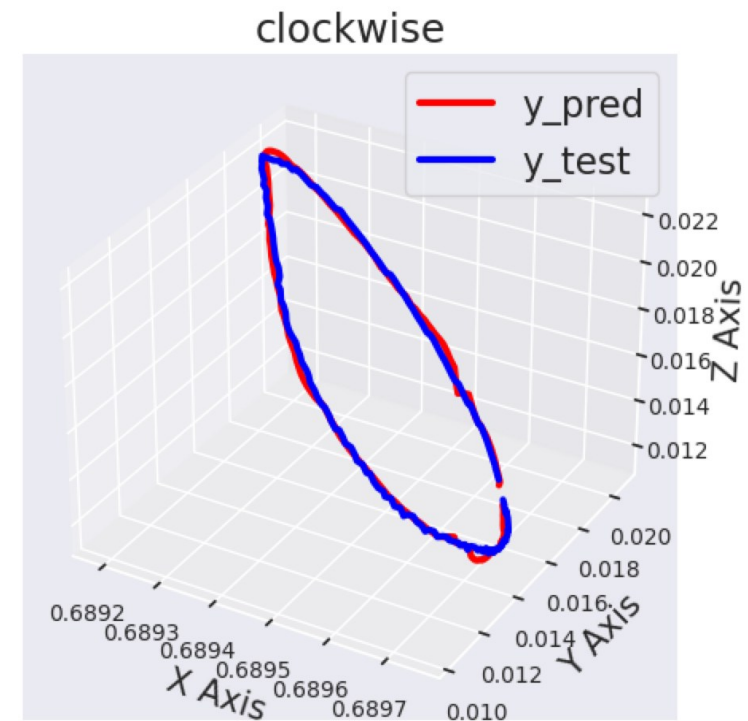
Dataset

- **[6] Endpoint (X/Y/Z):**
- This column shows the 3D coordinates of the end effector (the tip of the robotic arm).
 - The values inside this column are separated by slashes (/), meaning X, Y, and Z coordinates.
 - The origin (0,0,0) is located at the shoulder joint of the body
 - The unit is millimeters (mm).
 - Example: 392/-440/-84 in "circle/1.txt" → X = 392 mm, Y = -440 mm, Z = -84 mm.



Dataset

- You can see, the example movement and its trajectory is:
(never mind on the right-side graph scale!)



Dataset

- As of current, the trajectory set is limited to a small number (34 trajectories samples), but they are good samples.
 - (You might want to augment the data.)
- TA will upload the second chunk of data in the next week, which includes both good and “somehow noisy” data in 2nd round.
 - (Still, it would be a small number; and due to the noises, you need to consider augmentation technique again.)

Project Timeline

- Team meetup and Feedback based activities, starts on 11th week.
- The topic is well-settled, so no proposal presentation.
- Timeline
 - 11th week-14th week: Implementation
 - 15th week: Final Presentation with Demo run
 - 16th week: Documents deadline

Project Team

- Team buildup rules
 - No “forced” mix-up.
 - Three or four students can make a team.
- Mandates
 - Source codes and documentations should be uploaded on a repository (Github or similar)
 - The source codes or documentations should contain individual contributions.
 - Violations on academic integrity will results in an F.

What Will You Submit

- The model source code and running results.
- Final report – up to 10 pages.
 - Includes figures and tables,
 - Not includes references and cover page.
- Separately – Peer-review report
 - Will be announced in the last week; each teammate evaluate others.

Project Evaluation

- Evaluations
 - Dr. Young Hoon Ji (Hexar Humancare) will evaluate the projects.
 - Instructor's and peer-review will be counted.
- Team-based scoring and then,
 - Individual scores: (Team score) * Multiplying factor

Project Evaluation

- Implementation 70%
 - Code readability, model complexity and performance, etc.
 - All implementations must be interpretable and explainable.
- Documentation 30%
 - Thoroughness and details of the documents.
 - Always, explain “WHY” and “HOW” in your documentation.
 - Author Credit

Team buildup

- The team should let me and my TA know by email
 - Send the information to : chungminho@hanyang.ac.kr (our TA: Minho Chung)
 - Team name
 - Leader and members (with your student IDs).
- Deadline: tomorrow EOD. (23:59:59 KST, November 5th)

Academic Integrity

- I trust my students, but if any of violation has been spotted, it will be result in an immediate F and will be penalized by university's academic integrity affairs.